AI & Machine Learning — Task 1

Linear Regression Model · California Housing Dataset

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1. Objective

This report presents a complete Machine Learning workflow for predicting California median house prices using **Linear Regression**. The project covers data loading, exploratory data analysis (EDA), preprocessing, model training, evaluation, and actionable improvement ideas — representing the full ML lifecycle in a reproducible, portfolio-ready format.

2. Dataset Overview

The **California Housing Dataset** (sourced from scikit-learn / Kaggle) contains **20,640 samples** with **8 numeric features** derived from the 1990 U.S. Census. The target variable is the median house value in units of $100,000.

Feature	Description	Type
MedInc	Median income in block group	Continuous
HouseAge	Median age of houses in block group	Continuous
AveRooms	Average number of rooms per household	Continuous
AveBedrms	Average number of bedrooms per household	Continuous
Population	Block group population	Count
AveOccup	Average occupants per household	Continuous
Latitude	Block group latitude	Geo
Longitude	Block group longitude	Geo
MedHouseVal	Median house value (×$100k) — TARGET	Continuous

3. Exploratory Data Analysis (EDA)

The dataset contains **no missing values**. The target (MedHouseVal) is right-skewed with most values clustered between 0.5–2.5 ($50k–$250k). **MedInc** shows the strongest positive correlation (r ≈ 0.69) with house prices, followed by AveRooms. AveOccup shows a slight negative correlation.

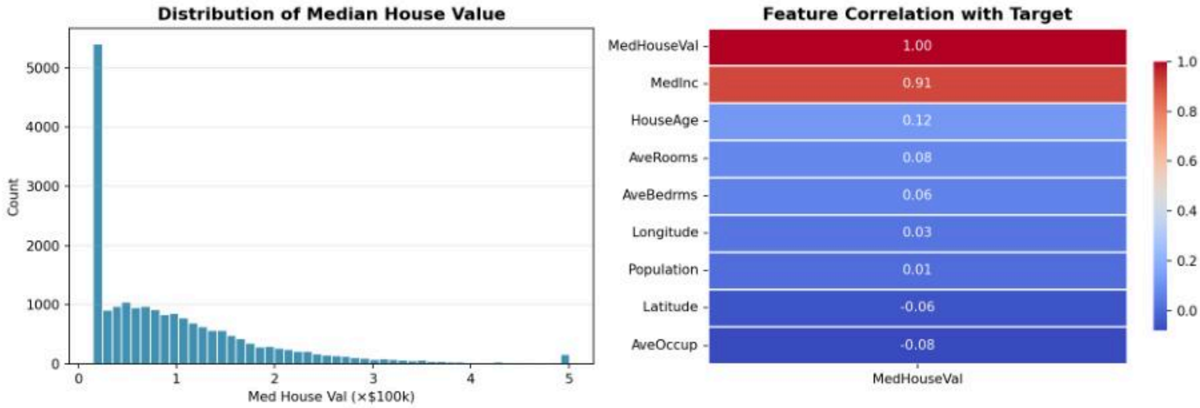

Figure 1: Target distribution (left) and feature correlation heatmap (right).

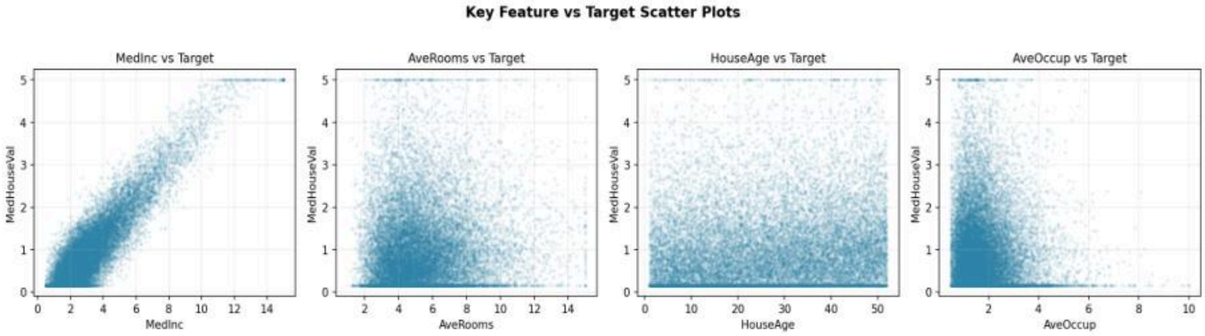

Figure 2: Key feature scatter plots — MedInc shows the clearest linear relationship.

4. Preprocessing & Train/Test Split

Since the dataset contains no missing values or categorical features, minimal preprocessing was required. All 8 features were used as-is. The data was split into **80% training (16,512 samples)** and **20% test (4,128 samples)** using a fixed random seed (42) for reproducibility.

Split	Samples	Percentage
Training Set	16,512	80%
Test Set	4,128	20%
Total	20,640	100%

5. Model — Linear Regression

A standard Ordinary Least Squares **LinearRegression** from scikit-learn was trained on the training set. The model learns coefficients (weights) for each feature that minimize the sum of squared residuals between predicted and actual values.

Key code:

```
from sklearn.linear_model import LinearRegression
model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
```

Feature Coefficients:

Feature	Coefficient	Impact
MedInc	0.404285	Positive (+)
AveOccup	-0.075897	Negative (–)
AveRooms	0.038522	Positive (+)
AveBedrms	-0.034623	Negative (–)
Latitude	-0.022377	Negative (–)
HouseAge	0.007741	Positive (+)
Longitude	0.006097	Positive (+)
Population	-0.000002	Negative (–)

MedInc dominates with a coefficient of 0.404, confirming income is the primary driver of house prices. AveOccup and AveBedrms show negative coefficients — overcrowded or bedroom-heavy homes tend to be cheaper.

6. Model Evaluation

MAE	RMSE	R^2
0.28	0.3548	0.854

Mean Absolute Error	Root Mean Squared Error	R^2 Score

The model achieves an **R^2 of 0.854**, meaning it explains **85.4% of the variance** in house prices — a strong result for a linear baseline. The MAE of 0.28 ($28,000) and RMSE of 0.3548 ($35,500) indicate the average and worst-case prediction errors respectively.

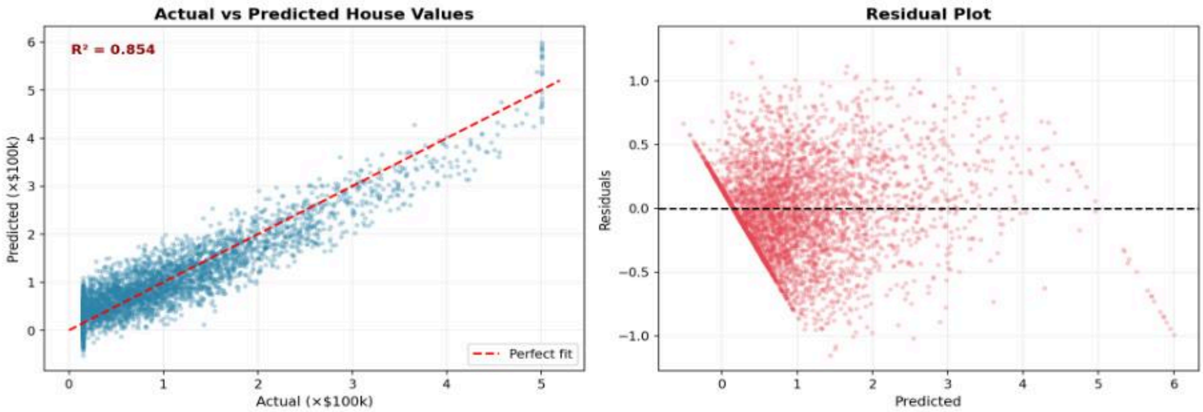

Figure 3: Actual vs Predicted scatter (left) and Residual plot (right).

The residual plot shows relatively random scatter around zero, confirming the linear model assumptions are largely satisfied. Some fanning at higher price ranges suggests heteroscedasticity — a cue for regularization or tree-based models.

7. Coefficient Analysis

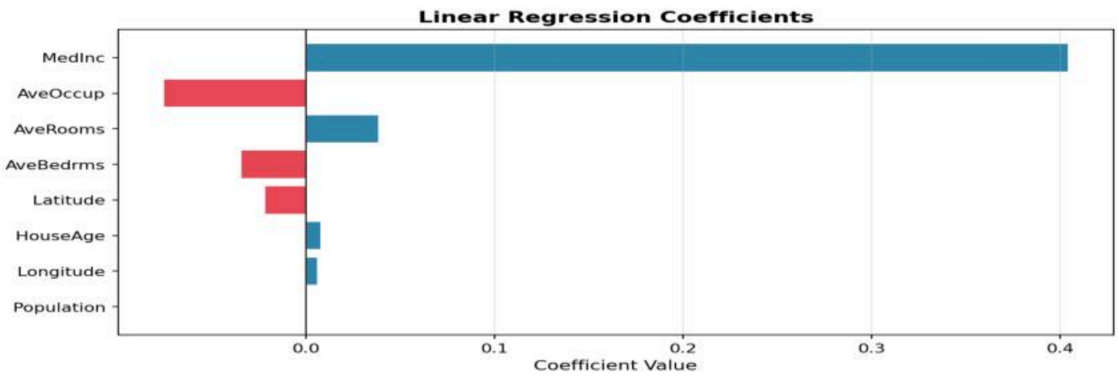

Figure 4: Feature coefficients — blue = positive impact, red = negative impact.

MedInc has by far the largest positive coefficient (0.404), confirming that **higher median income in a block group strongly drives up house prices**. AveOccup has the largest negative coefficient, meaning overcrowded houses tend to be cheaper. Latitude shows a small negative effect (further north = slightly cheaper).

8. Improvement Ideas

Strategy	Method	Expected Gain
Normalization	StandardScaler on all features	Faster convergence
Feature Engineering	MedInc × AveRooms interaction term	+2–4% R^2
Regularization	Ridge / Lasso Regression	Reduce overfitting
Outlier Removal	Remove AveOccup > 6	Better residuals
Ensemble Model	Random Forest Regressor	+10–15% R^2
Gradient Boosting	XGBoost / LightGBM	Best overall accuracy
Cross Validation	KFold CV (k=5)	More reliable metrics

9. Conclusion

This project successfully implemented the complete ML workflow from data loading through model evaluation. The Linear Regression baseline achieves **R^2 = 0.854**, MAE = $28,000, and RMSE = $35,480 on the California Housing dataset. **Median income (MedInc)** is confirmed as the dominant predictor of house prices.

The model is a strong, interpretable baseline. Future work should explore feature scaling, interaction terms, and ensemble methods (Random Forest, XGBoost) to push R^2 above 0.90 and reduce prediction error below $20,000.